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Key Points:
- Groundwater levels in the San Gabriel Valley Basin, California, reach all-time low after 2011-2016 drought
- Seismic velocities respond linearly with drawdown and recharge of ground water aquifer
- Change in seismic velocity change used to estimate change in groundwater storage

Supporting Information:
- Supporting Information S1

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Tracking Groundwater Levels Using the Ambient Seismic Field

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Abstract Aquifers are vital groundwater reservoirs for residential, agricultural, and industrial activities worldwide. Tracking their state with high temporal and spatial resolution is critical for water resource management at the regional scale yet is rarely achieved from a single type of groundwater data. Here we show that variations in groundwater levels can be mapped using perturbations in seismic velocity (*dv/v*). We measure *dv/v* in the San Gabriel Valley, California, from cross correlation of the ambient seismic field. *dv/v* reproduces the groundwater level changes that are marked by the multiyear depletions and rapid recharges, typical of California's cycles of droughts and floods. *dv/v* correlates spatially with vertical surface displacements and deformation measured with the Global Positioning System (GPS). Our results successfully predict the volume of water lost in the San Gabriel Valley during the 2012–2016 drought and thus provide a new, complementary approach to monitor groundwater storage.

Plain Language Summary The largest reservoir of freshwater worldwide is stored beneath our feet in groundwater aquifers. Monitoring the availability of water stored in the ground is critical for communities that are susceptible to drought, as groundwater can supplement a lack of surface fresh water. We propose a new way to monitor groundwater levels in aquifers using weak seismic waves repeatedly generated by the ocean, wind, and human activities. We exploit the fact that the speed of seismic waves is sensitive to the amount of the water in the ground. Using a network of seismometers, we measure the change in seismic velocity throughout the San Gabriel Valley Basin, California. We find that with the change in seismic velocity we can estimate the flux of water in the San Gabriel Valley over the last three droughts and subsequent recoveries that have affected Southern California over the last two decades. These results provide a new approach to monitor groundwater storage.

1. Introduction

Groundwater supplies one third of the fresh water used for residential, agricultural, and industrial use in the world (Döll et al., 2012). With increasing demand, overwithdrawal of groundwater has lead to subsidence and a loss of groundwater storage in numerous aquifers across the world (Galloway & Burbey, 2011). This is especially problematic for more than two billion people worldwide that live farther than 5 km from a source of surface fresh water (Kummu et al., 2011). Compounding this is the role of climate change, which has lead to more frequent and pronounced dry and hot years, along with stronger extreme precipitation events in places like California (Dettinger et al., 2011; Swain et al., 2016; Teng & Branstator, 2017). Accurate monitoring of groundwater levels is required to manage water supplies, yet comprehensive groundwater level measurements on the local to global scales are lacking (Wada et al., 2010), especially in the context of climate change (Taylor et al., 2012). Surface displacements measured by GPS provide high temporal but sparse spatial resolution of groundwater level changes (Bawden et al., 2001; Ji & Herring, 2012; King et al., 2007), while those measured by Interferometric Synthetic-Aperture Radar bring high spatial resolution but limited temporal resolution (Chaussard et al., 2017; Galloway & Hoffmann, 2007; King et al., 2007). Gravity measurements from the Gravity Recovery and Climate Experiment (GRACE) satellite are sensitive to water mass changes but only at large wavelengths, and they suffer nonuniqueness between water mass and aquifer depths (Rodell et al., 2009; Xiao et al., 2017).

The cross correlation of ambient seismic time series (Campillo, 2006; Shapiro & Campillo, 2004) can remedy these issues by providing high spatial and temporal resolution of the change in seismic velocity, *dv/v*, within a groundwater basin. *dv/v* has been widely used in recent years to study the dynamics of Earth's crust, for instance in response to earthquakes (e.g., Brenguier, Campillo, et al., 2008; Taira et al., 2015; Wegler et al., 2009), volcanic eruptions (Brenguier et al., 2016), and ice sheet melt (Mordret et al., 2016). Seasonal variations

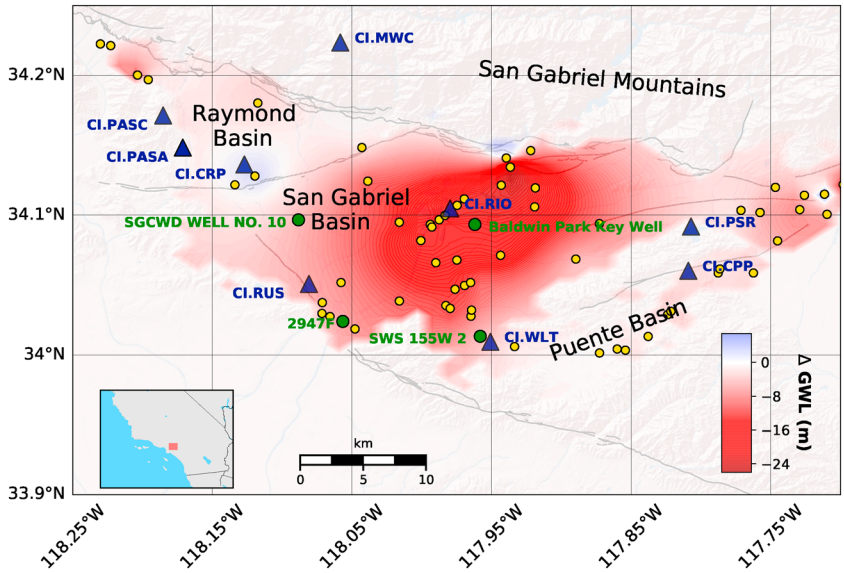

Figure 1. Groundwater level change in San Gabriel Valley during most recent drought (Fall 2012–Fall 2016). Seismic stations are shown as blue triangles, groundwater wells are shown as yellow circles, and *key* groundwater wells are shown as green circles.

from precipitation (Meier et al., 2010; Obermann et al., 2014; Sens-Schönfelder & Wegler, 2006; Tsai, 2011; Wang et al., 2017), air temperature changes (Hillers et al., 2015; Meier et al., 2010; Tsai, 2011), freeze-thaw of permafrost (Gassenmeier et al., 2015; James et al., 2017), and long-term variations from climatic forcing (Lecocq et al., 2017) are known to influence shallow seismic wave speeds. *dv/v* is conducive to monitoring groundwater levels, as the velocity of scattered seismic waves within water-saturated rocks is sensitive to changes in pore pressure (Grêt et al., 2006). Increasing pore pressure opens cracks and decreases the area of grain contacts, which in turns decreases seismic velocity (Christensen & Wang, 1985). Seismic waves naturally scatter in the Earth and provide an averaged and volumetric sampling of the medium. This contrasts with measurements from a ground water well, which are sensitive to a specific location (Healy & Cook, 2002) that may not be representative of the aquifer if the permeability structure is heterogeneous.

This study presents the perturbations in seismic velocity (*dv/v*) in the San Gabriel Valley (SGV), Eastern Los Angeles County, California. The SGV contains three unconfined, urban aquifers: the San Gabriel, the Puente, and the Raymond Basins. The east-northeast striking Raymond Fault acts as a barrier to flow between the Raymond and San Gabriel Basins, while the San Gabriel and Puente Basins are hydraulically connected (California Department of Water Resources, 1966; Main San Gabriel Watermaster, 2017; Yeats, 2004). Water-bearing sediments reach a maximum thickness of 1,200 m in the central part of the SGV (California Department of Water Resources, 1966). The SGV Basin is recharged by a combination of infiltration from rainfall, runoff from the San Gabriel Mountains, storm water capture, and imported water from the State Water Project. During droughts, groundwater supplies over 40% of water demand in the SGV (Main San Gabriel Watermaster, 2017). We consider changes in SGV groundwater levels from January 2000 to July 2017. This period is notable for having three major droughts in southern California (2002–2004, 2007–2009, and 2012–2016; Jones, 2015). During the most recent drought, groundwater levels dropped 18 m in the SGV in the Baldwin Park Key Well (BPKW; Figure 1), reaching all-time low levels in October 2016. Even with above average precipitation in the winter of 2016–2017, groundwater levels have only recovered 1.7 m in the SGV basin due to uptake by drought-parched soil (Main San Gabriel Watermaster, 2017).

2. Data and Methods

2.1. Ambient Seismic Noise Cross Correlation

We use continuous data from broadband vertical component seismometers in the California Integrated Seismic Network (CI) from January 2000 to July 2017 (Figure 1). To remove earthquakes and spurious signals, we discard hourly windows of raw data with maximum amplitude greater than 10 times the standard deviation of the daily value (Prieto et al., 2011). Raw waveforms are downsampled to 20 Hz, demeaned, and detrended. We then apply one-bit normalization and whiten in the frequency domain from 0.05 to 4 Hz

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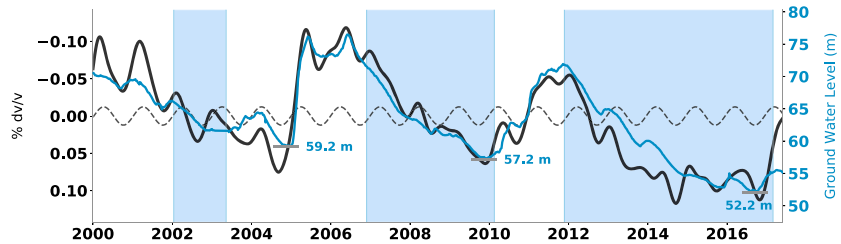

Figure 2. Relating seismic wave speed temporal perturbation to ground water levels. Observed *dv/v* stacked over all station pairs (black) with modeled *dv/v* due to thermoelastic strain (dashed) removed compared with groundwater change (blue) in the Baldwin Park Key Well. Grey bars indicate lowest historical water levels of the Baldwin Park Key Well. Blue patches indicate times of drought.

(Bensen et al., 2007; Lecocq et al., 2017). Noise cross-correlation functions (NCFs) are computed between all station pairs in 1-hr windows, with 30 min of overlap between successive windows, using the MSNoise package (Lecocq et al., 2014). Instrument corrections are applied after cross correlating. Daily NCFs for each station pair are formed by stacking all hourly NCFs. We use 30-day stacks of NCFs, as this is the minimum sampling period of our hydrologic data. The reference NCF for each station pair is the stack of all NCFs in the study period.

Daily changes in seismic velocity are computed using the Moving Window Cross-Spectrum (MWCS) technique (Clarke et al., 2011; Poupinet et al., 1984). MWCS computes time shifts, *dt*, between windows of data in daily NCFs relative to the reference NCF in the frequency domain. We calculate time shifts, using MWCS, beginning after the 0.5 km/s ballistic arrival in the NCFs and ending 50 s later. This region of the NCF, the *coda*, contains reflected and scattered waves, which are more sensitive to velocity changes than ballistic waves (Sens-Schönfelder & Wegler, 2006; Snieder et al., 2002). We use 10-s windows for MWCS and shift each window by 2 s (20% of the window length). We accept time shifts with $dt \leq 0.2$ s in each window and coherency $c \geq 0.5$ between the 30-day stack of NCFs and the reference NCF. A daily relative time shift dt/t is measured by a weighted linear regression of time shifts *dt* from each window in the causal and acausal part of the coda. Assuming that there is linear relation between relative time lags and that the velocity change is homogeneous throughout the sampling medium, the daily velocity variation is just $-dt/t = dv/v$.

2.2. *dv/v* Regionalization

Given that the SGV aquifer is an urban aquifer with localized groundwater extraction and recharge and heterogeneous hydrological properties, we expect velocity variations to be spatially localized. We use the regionalization method of Brenguier, Shapiro, et al. (2008) to map *dv/v* within our study area. This method approximates the sensitivity kernel of each station pair to a local velocity change as a small area around the direct raypath. While more realistic, diffuse sensitivity kernels based on single and multiple scattering are available (Margerin et al., 2016; Obermann et al., 2013), a scattering-based sensitivity kernel would have to account for 3-D velocity structure of the SGV basin. We use the direct raypath kernel for its simplicity, using a distance of 3 km from the direct raypath as the sensitivity of each station pair. We divide our study area into $50 \times 50 = 2,500$ grid cells (1,220 m × 905 m). We estimate the change in *dv/v* within each grid cell for every day in the study period as the average of all local station pair *dv/v* changes. Using this methodology, we can map *dv/v* changes spatially in the SGV through time. The *dv/v* regionalization shown in Figures 2 and 3 employs a 2-D Gaussian smoothing function.

2.3. Water Storage From *dv/v*

Coda wave observations have shown that *dv/v* can localize structural changes before volcanic eruptions (Brenguier, Shapiro, et al., 2008; Obermann et al., 2013). Here we use *dv/v* to estimate changes in groundwater volume, dV_w, within the SGV basin, over time. We form an empirical calibration constant, β, between groundwater level and *dv/v* changes in the SGV using groundwater levels recorded at the BPKW and the average *dv/v* change across all station pairs during the January to June 2005 rain event. We assume that β reflects the elastic properties of the aquifer, as King et al. (2007) reported that the uplift of the SGV during the recovery of groundwater levels in early 2005 was purely elastic. A 16.8-m increase in groundwater level in the BPKW and −0.00125 (−0.125%) change in *dv/v* for the SGV basin gives a value of $\beta = -13,280 \frac{m}{(\frac{m/s}{m/s})}$. We use our

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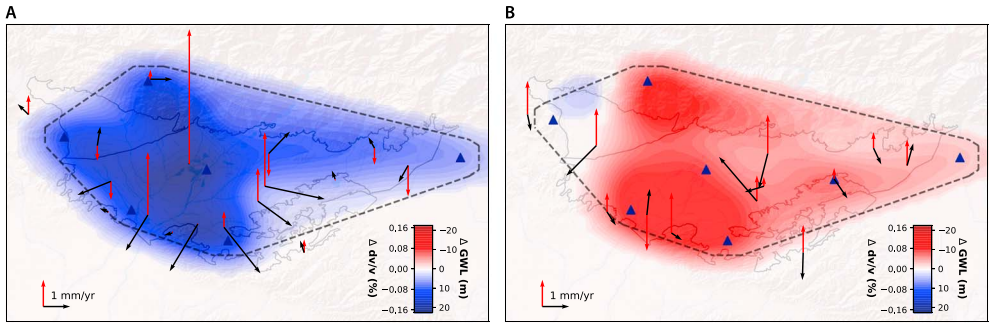

Figure 3. Regionalization of *dv/v*. (a) *dv/v* changes January 2005 to June 2005 following large precipitation event in the San Gabriel Valley. GPS stations (red = vertical, black = horizontal) uplift and move away from center of aquifer. (b) *dv/v* changes January 2012 to January 2017 during California's drought. GPS stations move toward center of the aquifer. The dashed black lines indicate extent of ray coverage. Scaling of *dv/v* and groundwater level is from empirical calibration of the winter 2004–2005 rain event.

calibration factor β at the BPKW to transform the *dv/v* change in each grid cell, $\frac{\Delta v}{v}$, to an equivalent relative change in groundwater, $dh = \frac{\Delta v}{v}\beta$. We calculate the relative in groundwater storage dV_w in each grid cell using

$$dV_w = S_y A \frac{\Delta v}{v} \beta \tag{1}$$

where S_y is the specific yield, A is the area of a grid cell in the regionalization of *dv/v*, and $\frac{\Delta v}{v}$ is the change in seismic velocity in a grid cell between two dates (Fitts, 2013). The specific yield, S_y, of an aquifer relates the volume of water released from storage per unit surface area of aquifer per unit decline in head (Fitts, 2013). S_y varies from 0.03 to 0.24 across the SGV, with averages of 0.14, 0.08, and 0.09 in the central, eastern, and western parts of the SGV, respectively (California Department of Water Resources, 1966). We take $S_y = 0.12$ as a representative, average value for the entire SGV basin. We integrate dV_w over all grid cells within the SGV to get a total relative water volume change within the basin.

3. Results and Discussion

We measure *dv/v* in the 0.5- to 2.0-Hz frequency range, which is greatly sensitive to the upper 500 m of the basin (see supporting information for surface wave sensitivity kernels; Herrmann, 2013). This frequency range is the most promising for groundwater monitoring in the SGV, where the depth to the water table is between 40 and 60 m below the surface. The change in groundwater level in the BPKW explains most of the variance in the evolution of *dv/v* in the SGV. We observe three distinct functional forms in our *dv/v* measurements: (1) seasonal periodicity, (2) impulsive events, and (3) multiyear linear trends (Figure 4).

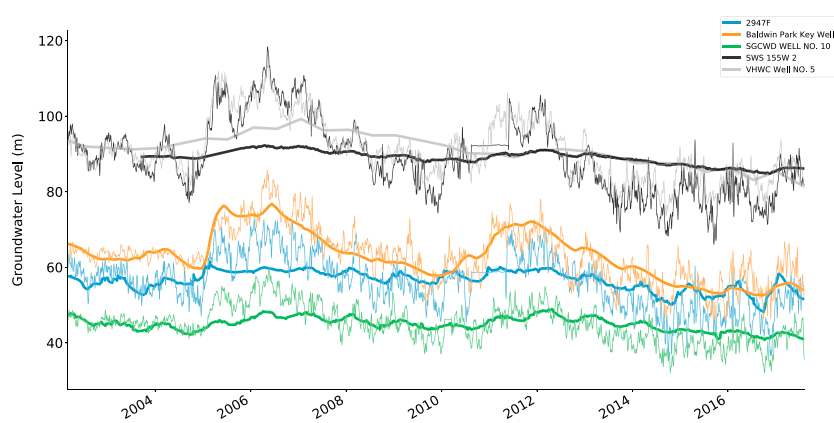

Figure 4. Groundwater levels at five key wells (thick line) in the San Gabriel Valley and relative groundwater level changes derived from regionalization of *dv/v* (thin line) at location of each well. Relative groundwater estimates from *dv/v* regionalization are scaled by calibration constant β and plotted at mean groundwater elevation for each well.

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Thermoelastic effects have been observed to dominate seasonal velocity changes in arid regions (Gassenmeier et al., 2016; Hillers et al., 2015; Richter et al., 2014). In the greater Los Angeles basin, which contains the SGV, Meier et al. (2010) were not able to distinguish between seasonal variations due to hydrological or thermoelastic effects. We correct for the best fitting seasonal signal in *dv/v* that is within the range of those expected for thermoelastic stresses (see supporting information for details of model). We find that seasonal thermoelastic strains induce perturbations in wave speed of about 0.02%, much lower than the hydrological effects that perturb elastic wave speeds to a of maximum 0.15%. *dv/v* lags behind surface temperatures in the SGV by $t = 94$ days. This high lag value is expected for an area that has unconsolidated sediment at the surface (Tsai, 2011). A component of the seasonal signal in *dv/v* we measure is likely the seasonal recharge in groundwater within the SGV basin (Jasechko et al., 2014).

At the end of 2004, groundwater levels were at an all-time low in the BPKW since measurements began in 1932. Groundwater levels rebounded in the winter of 2004–2005 due to 1 m of total precipitation, the highest amount of rainfall in 100-years in Los Angeles. Water levels in the BPKW increased by over 16 m in a span of 5 months. GPS stations recorded more than 40 mm of uplift in the central part of the SGV (Ji & Herring, 2012; King et al., 2007). We find that *dv/v* decreased by 0.15% in the same time frame. This drop in *dv/v* is similar in amplitude to that seen after nearby earthquakes (Brenguier, Campillo, et al., 2008; Wegler et al., 2009). The largest decrease in *dv/v* occurred in the center of the SGV (Figure 2a), where the basin is deepest (Yeats, 2004), as were the largest deformations recorded with Interferometric Synthetic-Aperture Radar (King et al., 2007). There is no statistically significant phase lag between the groundwater levels and *dv/v* response, suggesting a purely elastic response of the aquifer.

dv/v due to groundwater level changes have been observed in previous studies (Gassenmeier et al., 2015; Sens-Schönfelder & Wegler, 2006) but not in the context of water resource management or drought. Recently, Lecocq et al. (2017) found that thermoelastic strains, including those induced by long-term warming, and hydrologic loading contributed equally to *dv/v* over 30 years, due to relatively low fluctuations in groundwater levels, in the Grafenburg region, Germany. Here the dominant process that impacts the variations in elastic wave speed is the multiyear drawdown during periods of low precipitation (Teng & Branstator, 2017). During the drought of 2012–2016, groundwater levels declined in the SGV at a rate of 450 mm/year, which is one of the highest rates seen globally (Wada et al., 2010). During the period January 2012 to January 2017, when additional well data are available throughout the SGV, we find spatial and temporal correlation between the change in *dv/v* and groundwater levels. The largest change in *dv/v* during the drought occurred at two stations located within the basin and atop the thickest part of the aquifer (Figure 2b; California Department of Water Resources, 1966). A small decrease in *dv/v* in part of the Raymond basin during the same time frame suggests that the SGV and Raymond basins are hydraulically separated (California Department of Water Resources, 1966; Ji & Herring, 2012). GPS stations during the same time period measured a contraction of the ground surface that may result from a elastic response of the basin.

The regionalization of *dv/v* gives an intermediate picture of groundwater levels between the scale of an individual well to an entire basin. We extract relative groundwater level time series from the regionalization of *dv/v* by scaling *dv/v* at each location by β, the instantaneous elastic response of the 2005 rainfall event at the BPKW. *dv/v* reconstructions of groundwater level match well with measured levels at the BPKW and two key wells in western parts of the SGV, while reconstructions at the two wells at the eastern edge of the SGV are overestimated (Figure 3). The inaccuracy in the reconstructed groundwater level may arise from the lack of knowledge in the 3-D sensitivity of our measurement, in particular at the edge of the basin (see supporting information for *S* wave velocity map). Our regionalization scheme is also limited by the sparsity of the local seismic network. The strong temporal correlation between groundwater levels and *dv/v* in the center of the SGV, which is the area of the largest water level variations, provides us confidence to estimate changes in groundwater storage at key time periods. Applying our methodology in section 2.3 to the regionalization of *dv/v* from January 2012 to January 2017 yields an absolute water storage loss of 0.48 km^3. This matches well with the 0.45–0.5 km^3 of water that the Main San Gabriel Watermaster pumped from the SGV Basin during the drought to meet water demand (Main San Gabriel Watermaster, 2017).

Our results imply that the change in seismic velocity, *dv/v*, has tremendous potential to complement monitoring of groundwater fluctuations in basins of moderate-size aquifers. It has the capability to provide a direct and continuous monitoring of the spatial variations in groundwater levels, due to short-term recharge and long-term drawdown. Our analysis monitors groundwater fluctuations at an intermediary spatial resolution

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between monitoring wells and GRACE data. Future groundwater monitoring efforts with *dv/v* could attain higher spatial resolution of groundwater fluctuations with the use of a dense seismic network and more realistic sensitivity kernels.

Acknowledgments
Waveform data were accessed from the Southern California Earthquake Data Center (https://doi.org/10.7914/SN/CI). Groundwater data were accessed from the Los Angeles County Department of Public Works (http://ladpw.org/wrd/) and San Gabriel Watermaster (http://www.watermaster.org/). GPS time series data were accessed from NASA's Crustal Dynamics Data Information System (ftp://cddis.gsfc.nasa.gov/pub/GPS_Explorer/latest/). Weather data were accessed from the Western Regional Climate Center (https://wrcc.dri.edu/). Surface wave sensitivity kernels were computing using data from the Southern California Earthquake Center Community Velocity Model CVM-S4.26.M01 (https://scec.usc.edu/scecpedia/CVM-S4). The ObsPy, PyASDF, MPI4Py, SciPy ecosystem, and MSNoise packages were used to process data. The computations in this paper were run on the Odyssey cluster supported by the FAS Division of Science, Research Computing Group at Harvard University. The authors would like to thank Loïc Viens, Aurelien Mordret, Michel Campillo, Florent Brenguier, Victor Tsai, and James Rice for thoughtful and helpful discussions. T. C. analyzed the data (seismic, ground water, and GPS), created the figures, and wrote the manuscript. M. D. funded the research and edited the manuscript. M. D. and T. C. designed the project.

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Erratum

In the originally published version of this article, the legends for Figures 2, 3, and 4 were attributed to the wrong figures. The legends have been corrected so that the legend previously attributed to Figure 2 is now the legend for Figure 3; the legend previously attributed to Figure 3 is now the legend for Figure 4, and the legend previously attributed to Figure 4 is now the legend for Figure 2. The present version may be considered the authoritative version of record.

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